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Natalia A. Agapova

ML Engineer

Education

Innopolis University — Innopolis, Russia — 2022 – 2026

- **Major:** BSc in Applied Artificial Intelligence
- **Programming Coursework:** Operating Systems, Distributed and Network Programming, Networks, Information Retrieval, Advanced JavaScript, Industrial C++
- **AI/ML Coursework:** Data Structures and Algorithms, Reinforcement Learning, Practical Machine Learning and Deep Learning, Introduction to Computer Vision, Explainable, Interpretable and Fair AI, Generative Artificial Intelligence, Natural Language Processing

Employment

Croissan Studio — **Apr 2025 to present**

Client Work — **ML Software Engineer** — *Apr 2026 to present*

- Work as the ML engineer in client teams delivering end-to-end AI products, alongside designers and software engineers.
- Fully fine-tuned transformer language models and adapted others through prompting for client products.
- Compared model versions on task metrics to choose what goes into the product.
- Trained and adapted generative image and talking-avatar models for client projects.
- **Leveraged knowledge** in Python and PyTorch, fine-tuning Hugging Face Transformers models with LoRA and prompt engineering for generative AI products.

NeuroArtist, Pocket NeuroArtist — **ML Engineer & Product Designer** — *Jul 2025 to Apr 2026*

- Reworked the core model behind NeuroArtist, with the freedom to explore and test my own approach.
- Improved generated-image quality for Pocket NeuroArtist through iterative testing, growing its user numbers by about 5.5%.
- Restructured the UX of the Croissan Studio website, increasing clicks on the "Contact us" button by 13%.
- Also led the NeuroArtist rebrand across physical and digital materials.
- **Leveraged knowledge** in Python and PyTorch, building generative-AI image models with Diffusers and prototyping in Figma.

NeuroArtist — **Software Engineer Intern** — *Apr 2025 to Jul 2025*

- Worked on the NeuroArtist website.
- Helped with ML tasks on the team – including tracking down a bug in the image upscaler that had been mistaken for a model issue – and was offered an ML Engineer role after the internship.
- Contributed to client projects, for example a tourism website and the Moscow 2030 event, etc.
- **Leveraged knowledge** in Python for ML debugging and JavaScript/React for the website, versioned with Git and GitHub.

- *Letter of recommendation from the studio's CEO* →

Asimov Lab (startup) — **Frontend Team Lead & ML Developer** — **Feb to Sep 2024**

- Led the frontend team to the MVP of an AI platform that generates tests for teachers and students.
- Built the MVP version of the website end to end.
- One of six ML developers behind the core model of the product.
- **Leveraged knowledge** in React and TypeScript for the frontend, and Python with scikit-learn for the model.

Freelance — **ML Engineer** — **2022 to present**

- Applied ML work for independent clients.
- Clients included Russian Post, Innopolis City Administration, Innopolis University, Lyceum Innopolis, and others.
- **Leveraged knowledge** in Python, PyTorch, and scikit-learn for client ML projects.

Personal Projects

Fair Resume Screening — NLP · Explainability · Debiasing · Published, first author

- Built a BERT model that sorts resumes into 9 IT job categories using real HeadHunter data, then audited it for bias and tested 39+ ways to fix it.
- Found that swapping just the applicant's city flipped the model's decision in 7.7% of cases (baseline accuracy 60.9%) – published this bias finding as first author.
- **Utilized:** Python, PyTorch, BERT, Captum, pandas.

Emotion Detection — Computer Vision · On-device Deployment

- Trained a CNN to recognize emotions from faces, reaching 58.7% accuracy (up to 60.8% with ensembling), then shrank it to run on-device with just 1.15 ms latency (Core ML, ONNX Runtime Web).
- **Utilized:** Python, PyTorch, Core ML, ONNX Runtime.

Personal Knowledge System — RAG · NLP · Embeddings

- Built a personal AI assistant that answers only from your own PDFs and always cites the exact file and page, powered by a local LLM (Llama 3.2) so nothing leaves your machine.
- **Utilized:** Python, SentenceTransformer, ChromaDB, Ollama.

Publications

Agapova, N. A.; Lukmanov, R. A. "Improving Fairness in AI-Powered Recruitment: An Interpretable Resume Screening System", 2026. doi:10.66693/mathai.1017

Kholodov, Y. A.; Salloum, H.; Agapova, N. A. "Quantum Annealing Approaches to Breaking RSA Encryption", *Voprosy kiberbezopasnosti*, 2025. doi:10.21681/2311-3456-2025-4-127-133

Skills

ML / AI: *(proficient)*: Python, PyTorch, BERT, Hugging Face Transformers, Diffusers, LoRA fine-tuning, generative AI, prompt engineering, LLMs (Ollama), Captum, model evaluation & debiasing, RAG (ChromaDB, SentenceTransformer) *(familiar)*: scikit-learn, pandas, OpenCV, MediaPipe, Core ML, ONNX Runtime
Tools: *(proficient)*: Figma, Tilda, Git/GitHub *(familiar)*: React, TypeScript
Languages: Russian C2 (native) · English C1+ · Ukrainian B1 · French A2 · Tatar A2 · Korean A1